# Bibliometric and Temporal Analysis

## Bibliometric and Temporal Analysis I

Descriptive statistics summarize the corpus along every available axis: counts by year, venue, and author; citation-count distributions; and author productivity. Temporal analysis fits the publication time series, reporting a compound annual growth rate of 6.76% across 2000–2023 (a span of 23 years), with a mean year-over-year growth rate of 38.9% and a doubling time of 2.1 years. The peak publication year is 2023 with 9 records.

## Growth Metrics I

The compound annual growth rate (CAGR) is computed as:

$$\text{CAGR} = \left( \frac{N_{\text{end}}}{N_{\text{start}}} \right)^{1/(\text{year span})} - 1$$

where $N_{\text{start}}$ is the publication count in the first year (2000) and $N_{\text{end}}$ is the count in the last year (2023). The mean year-over-year growth rate $\bar{g}$ is the arithmetic mean of annual ratios. The doubling time is $t_d = \ln(2)/\ln(1 + \text{CAGR})$. These metrics are stored in `temporal_analysis.json` and injected into the manuscript at render time.

## Subfield Classification I

Subfield classification assigns each record to one of 4 configurable buckets (Observational Methods, Atmospheric Molecules, Jwst Instruments, and Theoretical Modeling) by priority-aware keyword matching; the taxonomy is defined entirely in configuration (`project_config.subfield_keywords`). The largest bucket is **Observational Methods** at 51.4% of the classified corpus. A per-subfield temporal breakdown (`subfield_timeline.json`) tracks how each sub-area has grown over time, enabling identification of emerging or declining research threads.

## Topic Modeling I

A TF-IDF term-weighting of titles and abstracts (Salton and Buckley 1988) feeds non-negative matrix factorization (NMF) (Lee and Seung 1999), implemented with scikit-learn (Pedregosa et al. 2011). NMF decomposes the document-term matrix $\mathbf{V} \approx \mathbf{WH}$, where $\mathbf{W}$ is the document-topic matrix and $\mathbf{H}$ is the topic-term matrix. The factorization extracts 5 latent topics that cross-cut the keyword taxonomy. The random seed is fixed at 42 for reproducibility. The reporting follows established systematic-review practice (Page et al. 2021), with every figure and statistic traceable to a committed artifact.